Readings:

- Networking Fundamentals
- RFC1918

Lecture 5: Network and Data Traffic Control

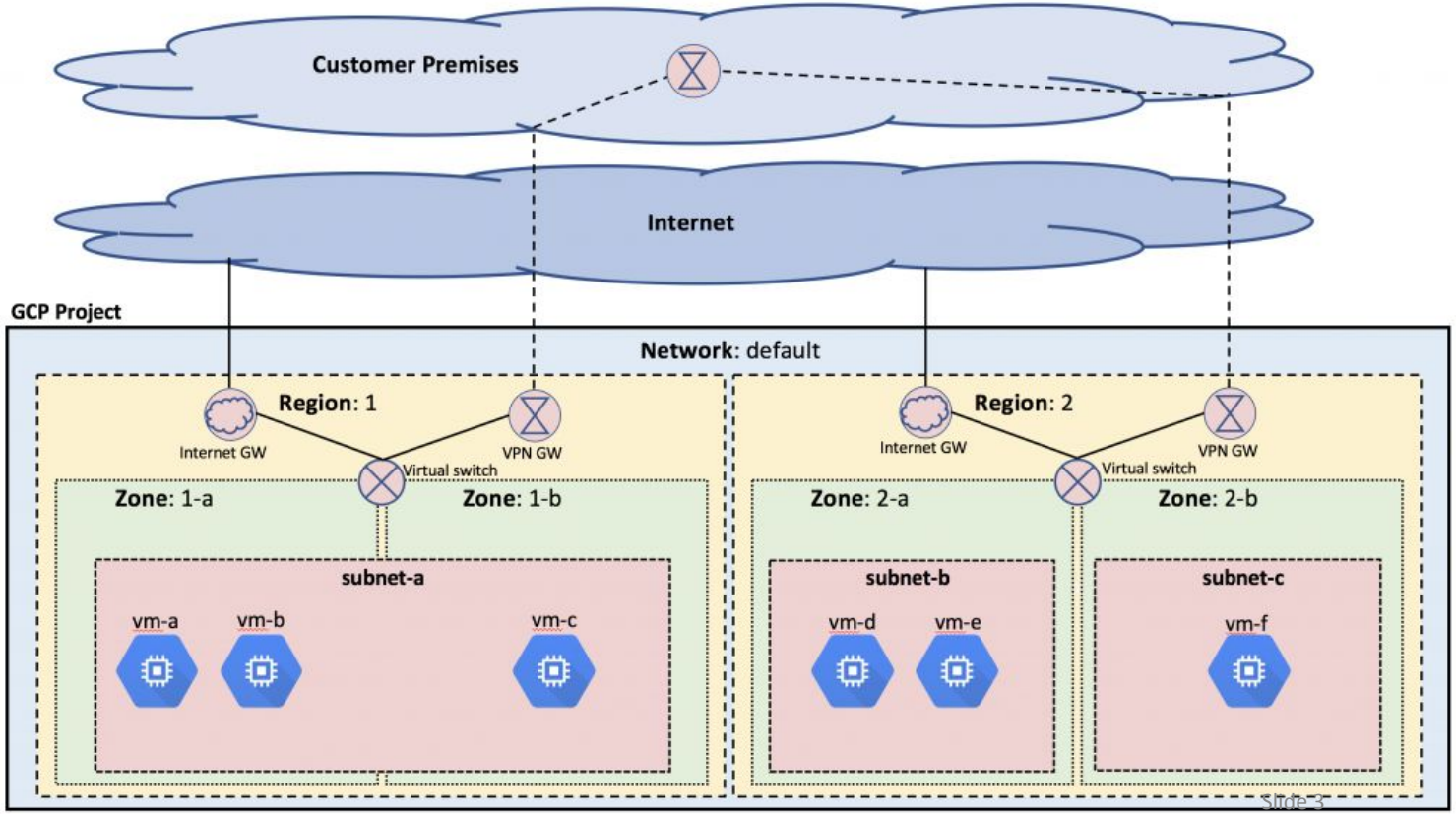
Lucerne University of
Applied Sciences and Arts
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Overview

- Regions and Zones
- Virtual Private Clouds
- Networks and Subnets
- IP Addressing Issues
- Routing
- Firewalls & Web Application Firewalls
- DNS Issues
- Dedicated Interconnect
- Network Billing

GCP Networking Basics

- VPCs
- Projects
- Networks
- Regions
- Zones
- Subnets
- Switching
- Routing
- Firewalls



https://www.networkmanagementsoftware.com/google-cloud-platform-gcp-networking-fundamentals/

AWS Regions

Regions

AWS has the concept of a Region, which is a physical location around the world where we cluster data centers. We call each group of logical data centers an Availability Zone. Each AWS Region consists of multiple, isolated, and physically separate AZ's within a geographic area. Unlike other cloud providers, who often define a region as a single data center, the multiple AZ design of every AWS Region offers advantages for customers. Each AZ has independent power, cooling, and physical security and is connected via redundant, ultra-low-latency networks. AWS customers focused on high availability can design their applications to run in multiple AZ's to achieve even greater fault-tolerance. AWS infrastructure Regions meet the highest levels of security, compliance, and data protection.

AWS provides a more extensive global footprint than any other cloud provider, and to support its global footprint and ensure customers are served across the world, AWS opens new Regions rapidly. AWS maintains multiple geographic Regions, including Regions in North America, South America, Europe, China, Asia Pacific, South Africa, and the Middle East.

https://aws.amazon.com/about-aws/global-infrastructure/regions_az/

AWS Availability Zones

Availability Zones

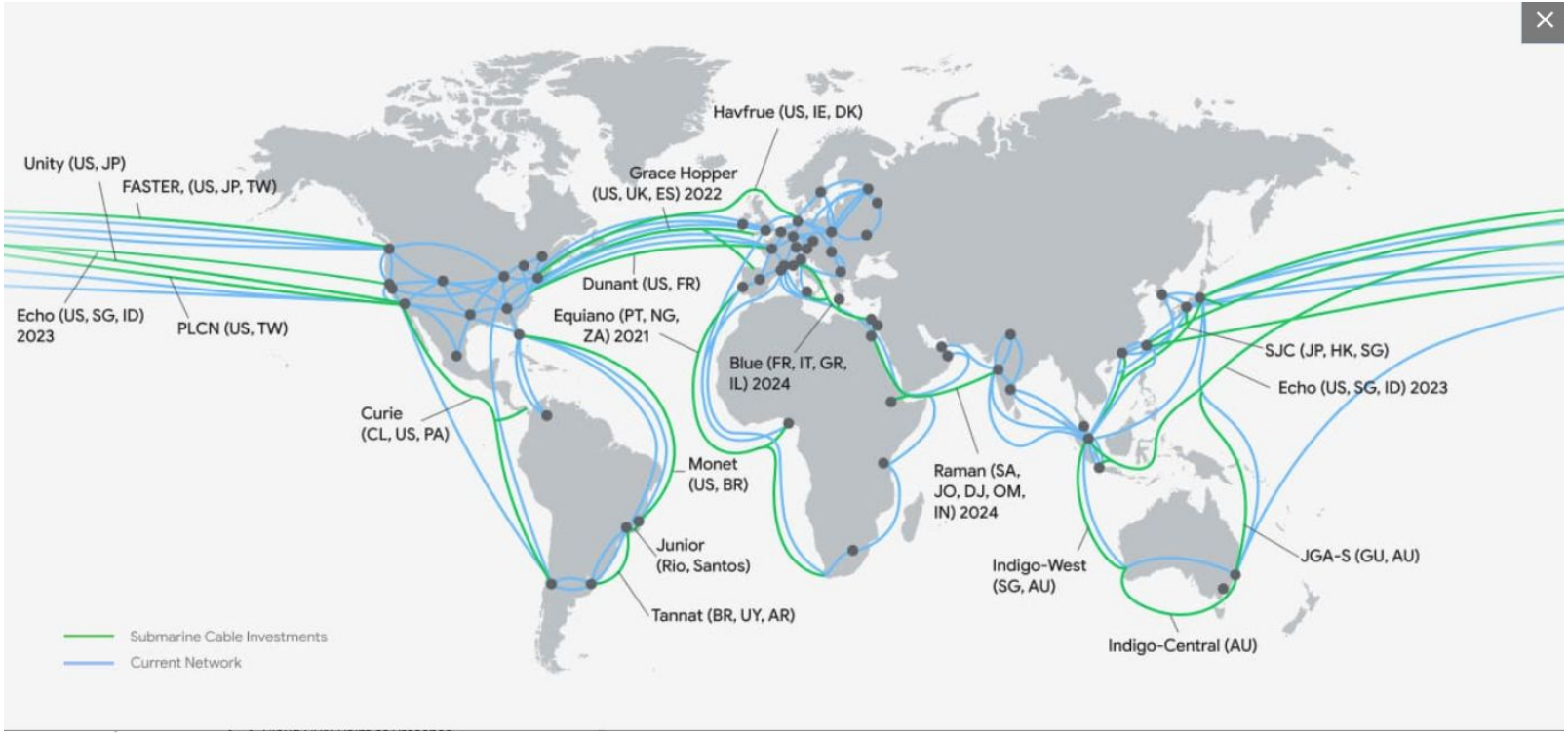
An Availability Zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity in an AWS Region. AZ's give customers the ability to operate production applications and databases that are more highly available, fault tolerant, and scalable than would be possible from a single data center. All AZ's in an AWS Region are interconnected with high-bandwidth, low-latency networking, over fully redundant, dedicated metro fiber providing high-throughput, low-latency networking between AZ's. All traffic between AZ's is encrypted. The network performance is sufficient to accomplish synchronous replication between AZ's. AZ's make partitioning applications for high availability easy. If an application is partitioned across AZ's, companies are better isolated and protected from issues such as power outages, lightning strikes, tornadoes, earthquakes, and more. AZ's are physically separated by a meaningful distance, many kilometers, from any other AZ, although all are within 100 km (60 miles) of each other.

https://aws.amazon.com/about-aws/global-infrastructure/regions_az/

GCP Regions


Oregon
Salt Lake City
Las Vegas
Los Angeles
Toronto
Montréal
Iowa*
Columbus
N Virginia
Dallas
S Carolina
Querétaro
Netherlands
Finland
London
Berlin
Belgium
Warsaw
Paris
Frankfurt
Turin
Zurich
Milan
Madrid
Tel Aviv
Dammam
Delhi
Doha
Mumbai
Seoul
Tokyo
Osaka
Taiwan
Hong Kong
Singapore
Jakarta
São Paulo
Santiago
Johannesburg
Sydney
Melbourne
Current region with 3 zones
Future region with 3 zones

GCP Global Network & Edge PoPs



https://cloud.google.com/about/locations#network

Cloud Region/Zone Comparison -- Exercise

EXERCISE FOR YOU TO COMPLETE!

	Regions	Zones
Alibaba Cloud		
AWS		
Azure		
Google Cloud		
Huawei Cloud		
Oracle Cloud		
Swisscom		
Tencent Cloud		

Example GCP Zones / CH

			Arm, Google Axion		
europe-west6-a	Zurich, Switzerland, Europe	E2, N2, N1, C2	Intel Ivy Bridge, Sandy Bridge, Haswell, Broadwell, Skylake, Cascade Lake, Ice Lake		Low CO_2
europe-west6-b	Zurich, Switzerland, Europe	E2, N4, N2, N2D, N1, C4, T2D, M3, M1, C2, G2, Z3	Intel Ivy Bridge, Sandy Bridge, Haswell, Broadwell, Skylake, Cascade Lake, Ice Lake, Sapphire Rapids, Emerald Rapids, AMD EPYC Milan	GPUs, AMD SEV	Low CO_2
europe-west6-c	Zurich, Switzerland, Europe	E2, N4, N2, N2D, N1, C4, T2D, M3, M1, C2, G2	Intel Ivy Bridge, Sandy Bridge, Haswell, Broadwell, Skylake, Cascade Lake, Ice Lake, Emerald Rapids, AMD EPYC Milan	GPUs, AMD SEV	Low CO_2

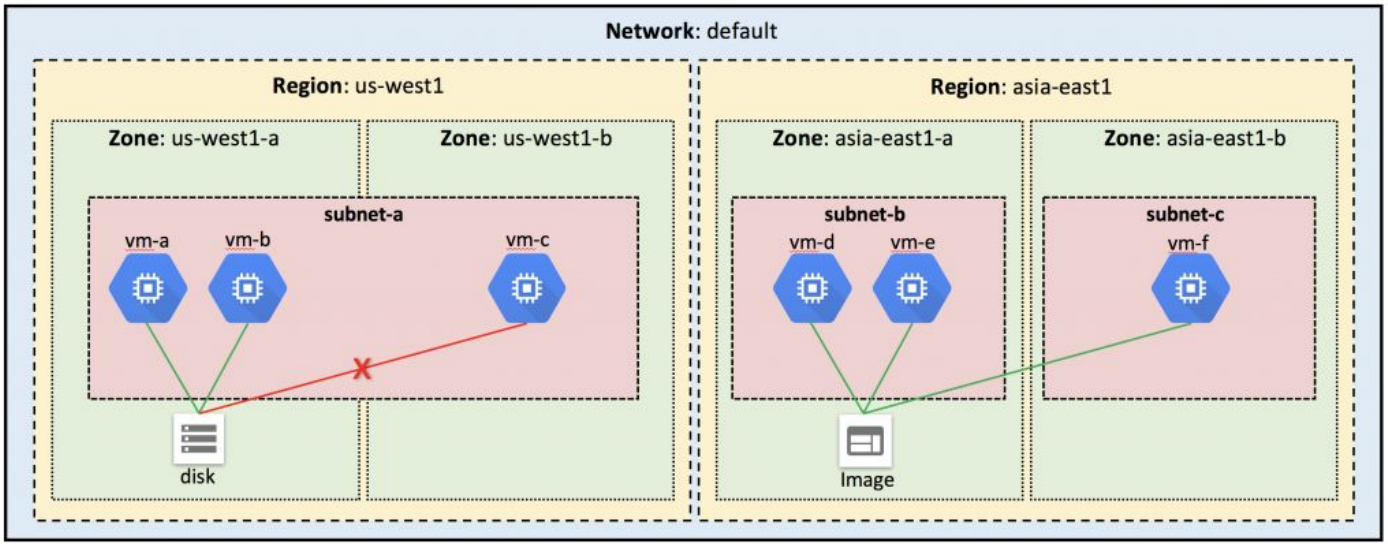
Example GCP Zones / Europe

Zones	Location	Machine types	CPUs	Resources
europe-north1-a	Hamina, Finland, Europe	E2, N1, M1	Broadwell, Skylake	
europe-north1-b	Hamina, Finland, Europe	E2, N1,	Broadwell, Skylake	
europe-north1-c	Hamina, Finland, Europe	E2, N1, M1	Broadwell, Skylake	
europe-west1-b	St. Ghislain, Belgium, Europe	E2, N2, N2D, N1, M1, C2	Sandy Bridge, Haswell, Broadwell, Skylake, Cascade Lake, AMD EPYC Rome	GPUs
europe-west1-c	St. Ghislain, Belgium, Europe	E2, N2, N2D, N1, C2	Ivy Bridge, Haswell, Broadwell, Skylake, Cascade Lake, AMD EPYC Rome	
europe-west1-d	St. Ghislain, Belgium, Europe	E2, N2, N2D, N1, M1, C2	Haswell, Broadwell, Skylake, Cascade Lake, AMD EPYC Rome	GPUs
europe-west2-a	London, England, Europe	E2, N2, N1, C2	Broadwell, Skylake, Cascade Lake	GPUs
europe-west2-b	London, England, Europe	E2, N2, N1, M1, C2	Broadwell, Skylake, Cascade Lake	GPUs
europe-west2-c	London, England, Europe	E2, N2, N1, M1, C2	Broadwell, Skylake, Cascade Lake	
europe-west3-a	Frankfurt, Germany Europe	E2, N2, N1, M1, C2	Broadwell, Skylake, Cascade Lake	
europe-west3-b	Frankfurt, Germany Europe	E2, N2, N1, M1, C2	Broadwell, Skylake, Cascade Lake	GPUs
europe-west3-c	Frankfurt, Germany Europe	E2, N1	Broadwell, Skylake	
europe-west4-a	Eemshaven, Netherlands, Europe	E2, N2, N1, C2	Broadwell, Skylake, Cascade Lake	GPUs
europe-west4-b	Eemshaven, Netherlands, Europe	E2, N2, N2D, N1, M2, M1, C2	Broadwell, Skylake, Cascade Lake, AMD EPYC Rome	GPUs
europe-west4-c	Eemshaven, Netherlands, Europe	E2, N2, N2D, N1, M2, M1, C2	Broadwell, Skylake, Cascade Lake, AMD EPYC Rome	GPUs
europe-west6-a europe-west6-b europe-west6-c	Zurich, Switzerland, Europe	E2, N1	Skylake	

Global vs Regional vs Zonal Resources

- While some of the core resources in GCP are global, others may be restricted by region or zone.
  - Regional resources can be used anywhere within the same **region**, while
  - Zonal resources can be used anywhere within the same **zone**.

- **Global** Resources:
  - Images
  - Snapshots
  - VPC Network
  - Firewalls
  - Routes
- **Regional** Resources:
  - Static external IP addresses
  - Subnets
- **Zonal** Resources:
  - Instances (VMs)
  - Persistent Disks



https://www.networkmanagementsoftware.com/google-cloud-platform-gcp-networking-fundamentals/

Advantages of Regions and Zones?

- To achieve *high availability* and *fault tolerance* for applications, resources can be distributed across multiple zones and regions.
- Zones are independent of each other, with completely separate physical infrastructure, networking and isolated control planes that ensure typical failure events only affect that zone.
- Applications should therefore be distributed across zones, to handle the failure of any particular zone. The same applies to regional issues.
- Another consideration for design is speed and latency. Zones have high-bandwidth, low-latency connections to other zones in the same region.
- If most user traffic will be initiated from certain parts of the globe, it's best to design for regions and zones closest to that point of service.

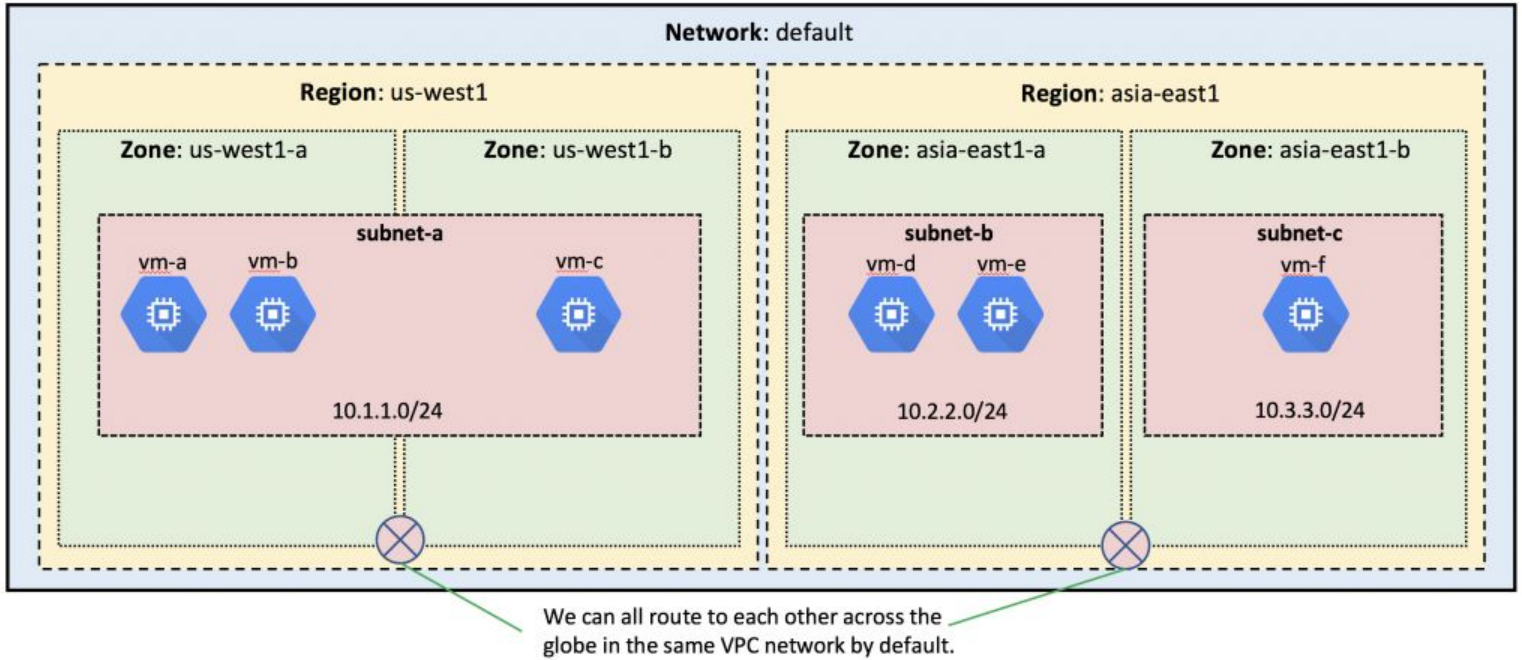
https://www.networkmanagementsoftware.com/google-cloud-platform-gcp-networking-fundamentals/

Virtual Private Cloud (VPC)

- "A Virtual Private Cloud (VPC) is a global private isolated virtual network partition that provides managed networking functionality for your Google Cloud Platform (GCP) resources." Google
- A VPC is like a virtual version of a traditional physical network.
- VPCs are global, spanning all regions.
- Instances within the VPC have internal IP addresses and can communicate privately with each other across the globe.
  - This logical representation of the network infrastructure abstracts much of the complexities of dealing with on-premises architectures.

https://www.networkmanagementsoftware.com/google-cloud-platform-gcp-networking-fundamentals/

VPC Default Global Routing



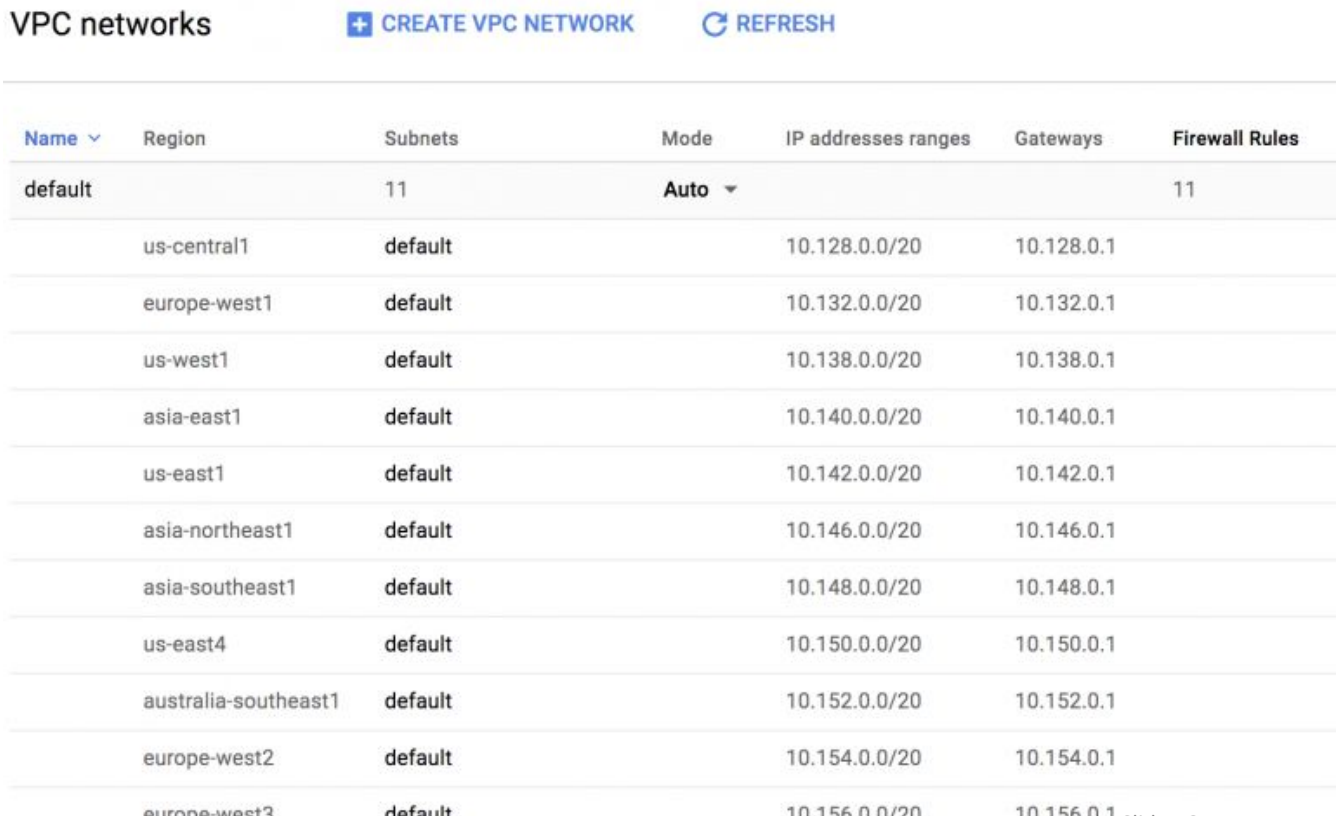
- The default VPC network spans multiple regions and zones, and subnets within various parts of the network servicing VMs.
- All of these subnets can natively route to each other, and as long as the firewalls permit it, VMs can reach one another within this VPC.

VPC Networks and Subnets

- When creating GCP projects the option is provided to automatically create a VPC network or custom build it.
- In *auto mode* – a default network with a series of default subnets is deployed ready for immediate use. VM instances can be deployed on the default subnets without any network configuration.
  - NB Do not confuse “VPC network” with something related to IP addresses. Instead, relate “VPC network” to that of a network topology or if you have a networking background – VRF.
- Each network has its own subnets, routes, firewall, internal DNS, and more beyond the basics listed here.
- GCP Project
  - VPC Network
    - Subnets
    - Routes
    - Firewall
    - Internal DNS

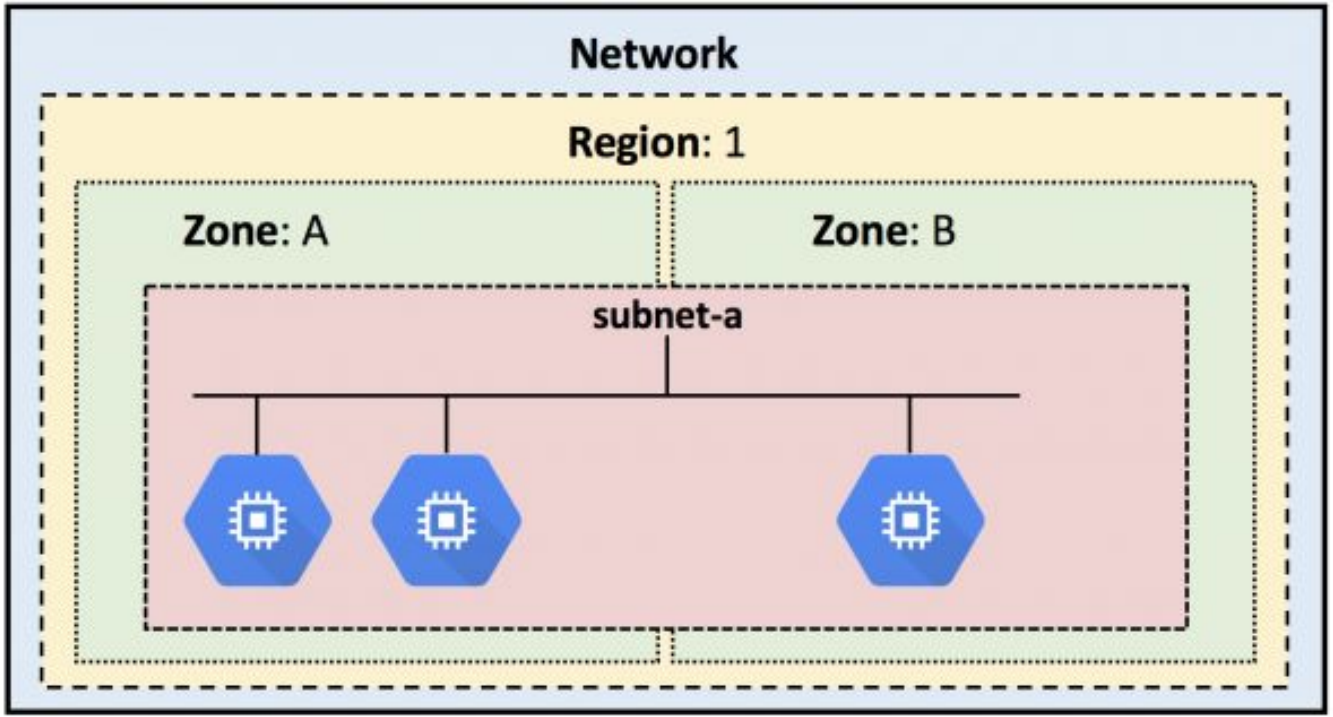
Screenshot from GCP console showing default network + a default subnet in each region

- The VPC network called *default* is **global** , while each of the subnets within it is **regional**.
- When you create an instance it is placed in a subnet.
- Instances in this subnet can be in any zone within the region it is assigned to.
- Even though subnets are regional, instances can communicate with other instances in the same VPC network using their private IP addresses.
- Subnets within the network can be isolated using firewall policies.



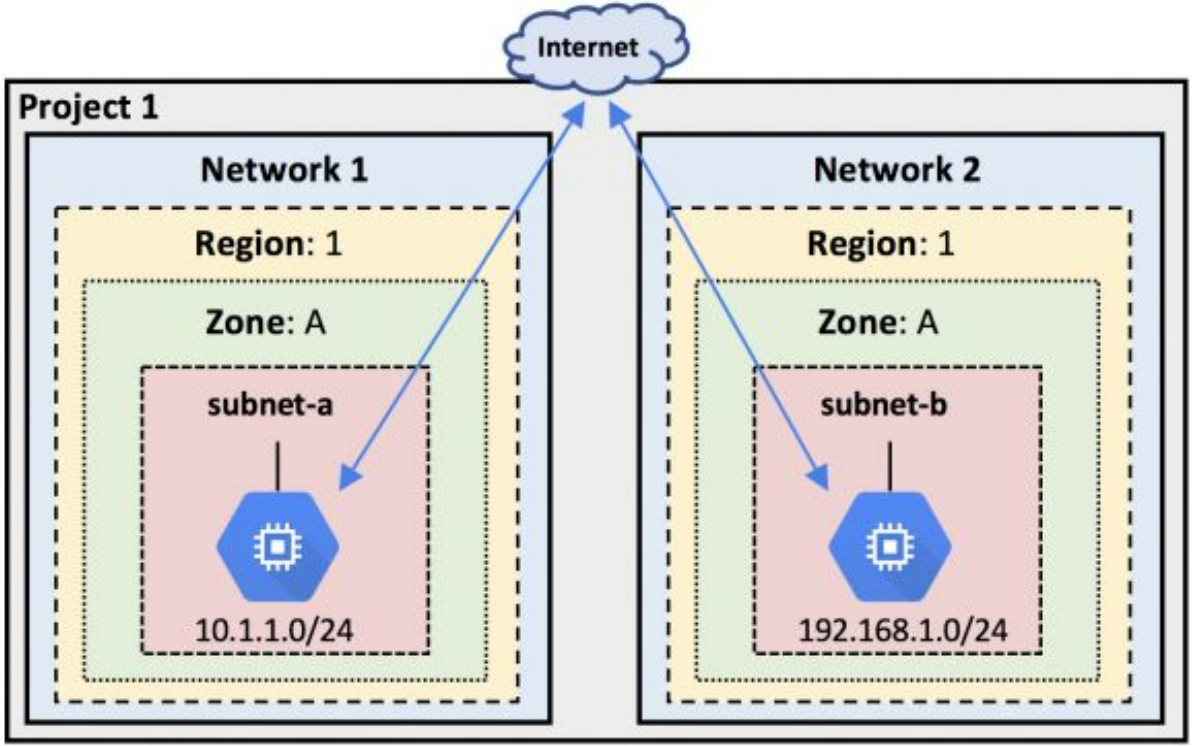
Name	Region	Subnets	Mode	IP addresses ranges	Gateways	Firewall Rules
default		11	Auto			11
	us-central1	default		10.128.0.0/20	10.128.0.1	
	europe-west1	default		10.132.0.0/20	10.132.0.1	
	us-west1	default		10.138.0.0/20	10.138.0.1	
	asia-east1	default		10.140.0.0/20	10.140.0.1	
	us-east1	default		10.142.0.0/20	10.142.0.1	
	asia-northeast1	default		10.146.0.0/20	10.146.0.1	
	asia-southeast1	default		10.148.0.0/20	10.148.0.1	
	us-east4	default		10.150.0.0/20	10.150.0.1	
	australia-southeast1	default		10.152.0.0/20	10.152.0.1	
	europe-west2	default		10.154.0.0/20	10.154.0.1	
	europe-west3	default		10.156.0.0/20	10.156.0.1	

Subnet Spanning Zones

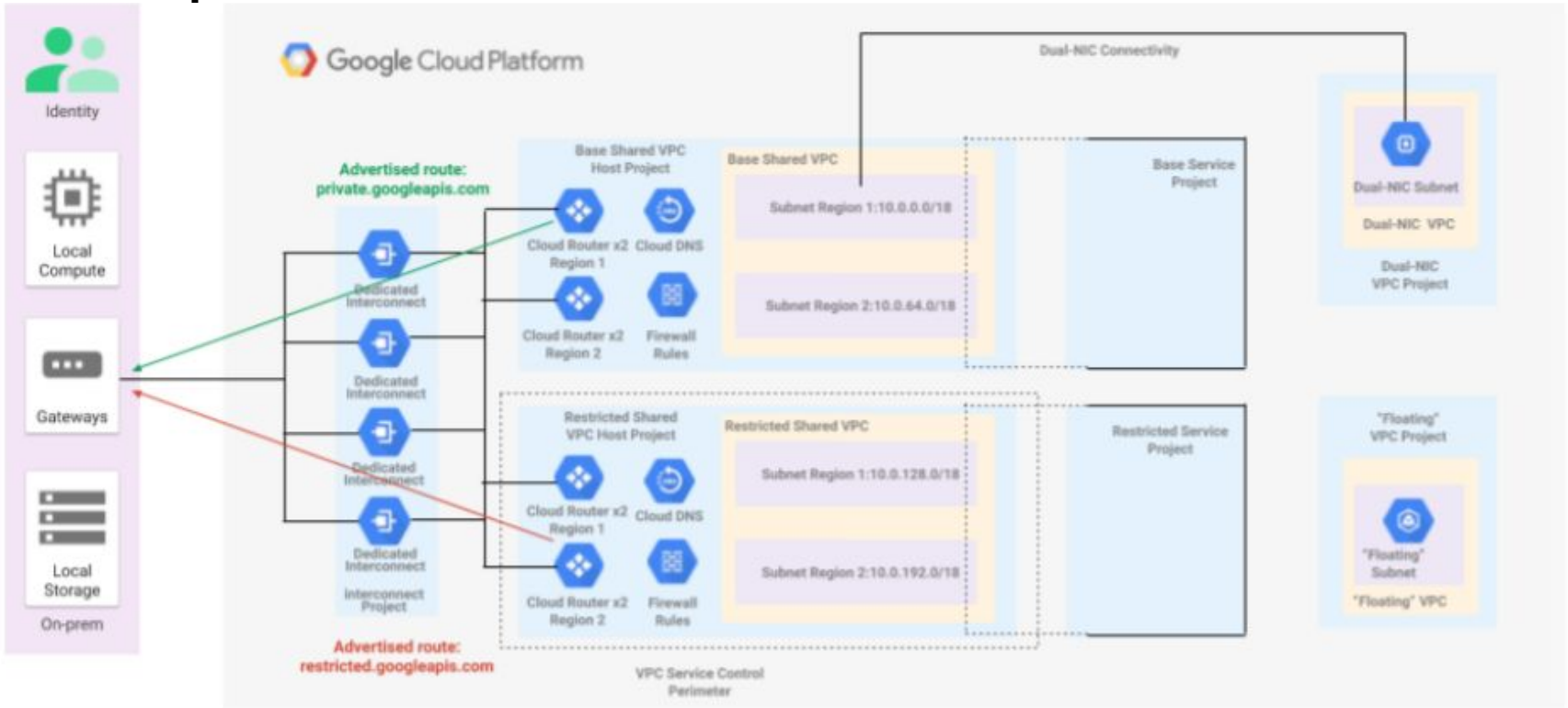


Separate Networks to Achieve Isolation

- For complete isolation between various applications, customers, etc., multiple networks can be created.
- Up to five networks per project can be created , including the default network.
- Multiple networks within a single project can provide multi-tenancy, IP overlap, or isolation within the project itself.
- This is an option instead of having multiple projects.



Example VPC Structure



Shared VPC Per Environment:
Prod/Non-prod//Dev

Quiz 9

- What is typically the difference between a Region and a Zone in cloud provider terminology?
- Find an example (or more!) of a Global, Regional and Zonal resource
- Can VPC subnets span zones in the GCP model?



IP Addresses

- Each VM instance in GCP has an internal IP address and typically an external IP address.
  - The internal IP address is used to communicate between instances in the same VPC network,
  - The external IP address is used to communicate with instances in other networks or the Internet.
  - These IP addresses are ephemeral by default but can be statically assigned.
- Internal IPs are allocated to instances from the subnet's IP range via DHCP. This means the IPs are ephemeral and will be released if the instance is deleted.
- External IPs are also assigned via DHCP from some Google-provided pool. These IPs are mapped to the internal IPs of the VM instances for you. You can reserve static IP addresses if needed.
- Regarding Static External IP Addresses, they can be either global or regional depending on your requirements. For example, global static IP addresses are available for global forwarding rules used for global load balancing.

https://www.networkmanagementsoftware.com/google-cloud-platform-gcp-networking-fundamentals/

IP Space Allocation

- To establish a Google Cloud foundation an IP address range must be allocated for the Google Cloud environment
- GCP supports a variety of valid IP ranges (both RFC1918 and non RFC1918)
- There are some reserved and restricted IP ranges which cannot be used in a subnet

Address Allocation for Private Networks / RFC1918

- The Internet Assigned Numbers Authority (IANA) has reserved the following three blocks of the IP address space for private internets:
  - 10.0.0.0 - 10.255.255.255 (**10/8** prefix) 24-bit block (a single class A network number, in pre CIDR notation)
  - 172.16.0.0 - 172.31.255.255 (**172.16/12** prefix) 20-bit block (16 contiguous class B network nums)
  - 192.168.0.0 - 192.168.255.255 (**192.168/16** prefix) 16-bit block (256 contiguous class C network nums)

Azure IP Addressing

CLOUD COMPUTING > IAAS/PAAS

Non-RFC 1918 IP Addresses in Azure Virtual Networks

John Savill | Nov 28, 2014

SEARCH LOG IN REGISTER NEWSLETTER SIGN-

Q: I've heard that I can now use non-RFC 1918 IP addresses in an Azure virtual network; what does this mean?

A: RFC 1918 specifies IP addresses that are assigned to be used internally (i.e., within an organization) and will not route on the Internet. Specifically, these are:

- 10.0.0.0/8
- 172.16.0.0/12
- 192.168.0.0/16

When you create an Azure virtual network, an IP address range is assigned that's then divided into virtual subnets. The IP address range selected prior to this update required the IP addresses to be from the RFC 1918 range (i.e., non-routable on the Internet). However, numerous customers had very large IP address spaces that were Internet-routable and they wanted to use those addresses in Azure, the same way they use them on-premises. The update to Azure virtual networks announced at TechEd Europe 2014 now allows virtual networks to use non-RFC 1918 IP address ranges. Of course, it's important to use non-RFC 1918 addresses only for IP spaces that you own in your organization. For most organizations, the use of RFC 1918 ranges is still the best option, and non-RFC 1918 use would be on an exception basis.

0 COMMENTS

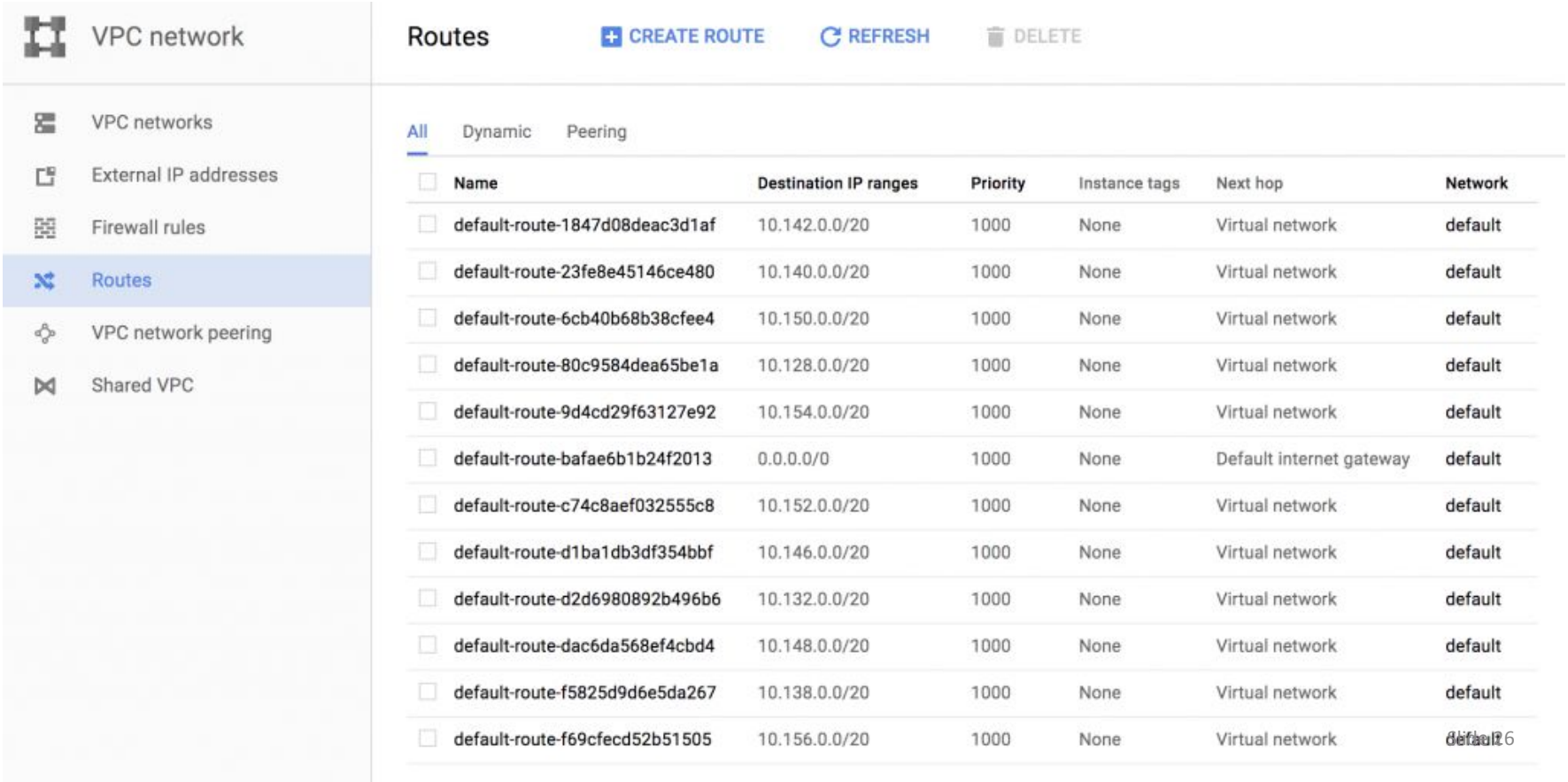
HOCHSCHULE LUZERN

https://www.itprotoday.com/iaaspaas/non-rfc-1918-ip-addresses-azure-virtual-networks

Routes

- GCP is a global software-defined network with layers of controllers and advanced routing capabilities that have been mostly abstracted for the end-user.
- All networks have routes in order to communicate with each other.
- The default network has a default route to the internet and individual routes to each subnet.
- On the following slide a screenshot of the GCP console is provided …

VPC Networks



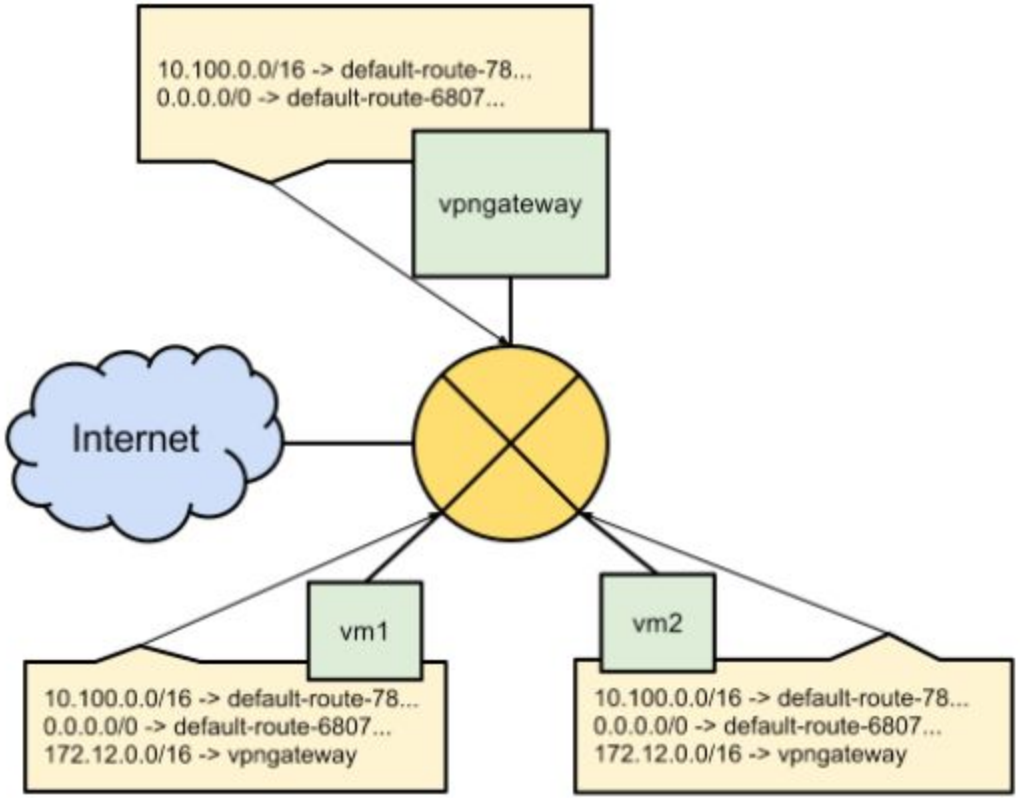
Routes

- Routes are considered a “network resource” and cannot be shared between projects or networks.
  - Tables are used to tell which routes and rules apply to each VM instance.
  - Routes could apply to multiple instances or single instances depending on the tags used in the route statement.
  - If an instance tag is used, the route applies to that instance, and if an instance tag is not used, then the route applies to all instances in that network.
  - Individual read-only route tables are created for each VM instance based on the parent route table.
- The following picture helps visualize the concept of these route tables. Even though there are no “routers” in the software-defined network, you can still think of each VM instance as connected to some core router, with all traffic passing through it based on the perspective of each node’s individual route table.

https://www.networkmanagementsoftware.com/google-cloud-platform-gcp-networking-fundamentals/

Traffic Routing

- Routing decisions apply to traffic egressing a VM. The most specific route in the table will match. Traffic must match a route and firewall rule in order for it to pass.
- You can add custom static routes or set up BGP for dynamic routing between clouds or on-premises environments.



https://www.networkmanagementsoftware.com/google-cloud-platform-gcp-networking-fundamentals/

Firewalls

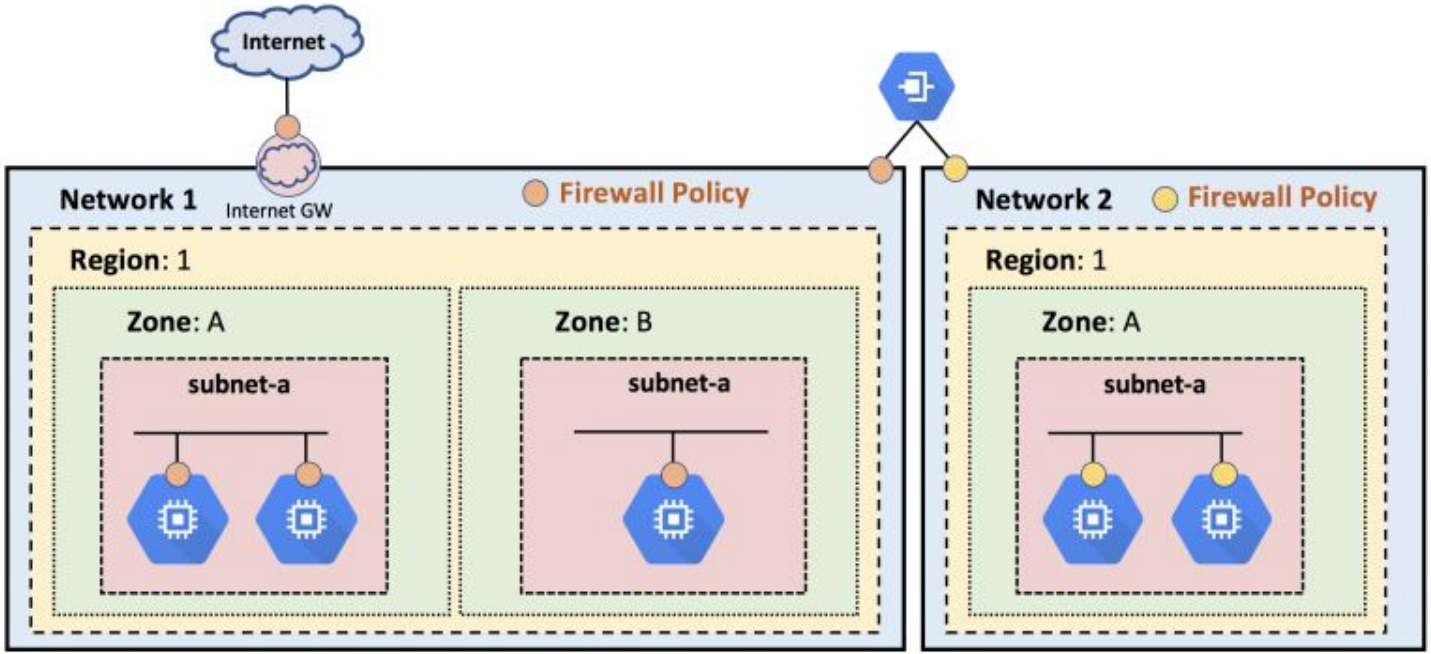
- Each VPC network has its own distributed firewall, which allows or denies traffic into and out of the network, and between VM instances.

- Note that traffic does not flow through some single firewall chokepoint device somewhere: GCP is a full SDN, with firewall policies applied at the instance-level, no matter where it resides. These checks are performed immediately without having to funnel traffic through dedicated security appliances.

https://www.networkmanagementsoftware.com/google-cloud-platform-gcp-networking-fundamentals/

GCP Firewall Rules

- Firewall rules can match IP addresses or ranges, but can also match **tags**.
- Tags are user-defined strings that help organize firewall policies for standards-based policy approach.
- E.g. you could have a tag called web-server, with a firewall policy that says any VM with the tag web-server should have ports HTTP, HTTPS, and SSH opened.
- Firewall rules are at network resource level and not shared between projects.



Google Cloud Firewall

- Fully stateful firewalls
- Fully distributed, no single choke point
- Ingress and egress access control
- Micro-segmentation with network tags and service accounts

- Firewall rules
  - Actions allow / deny
  - Apply to:
    - All VMs in VPC network or
    - A subset of VMs in the VPC, *target-tags* or service accounts
  - For more info see https://cloud.google.com/vpc/docs/using-firewalls

https://cloud.google.com/vpc/docs/firewalls#default_firewall_rules

VPC Network Implied Firewall Rules

Every VPC network has two implied firewall rules. These rules exist, but are not shown in the Cloud Console:

Implied allow egress rule.
An egress rule whose action is allow, destination is 0.0.0.0/0, and priority is the lowest possible (65535) lets any instance send traffic to any destination, except for traffic blocked by Google Cloud.
A higher priority firewall rule may restrict outbound access. Internet access is allowed if no other firewall rules deny outbound traffic and if the instance has an external IP address or uses a Cloud NAT instance.
For more information, see Internet access requirements.

Implied deny ingress rule.
An ingress rule whose action is deny, source is 0.0.0.0/0, and priority is the lowest possible (65535) protects all instances by blocking incoming connections to them.
A higher priority rule might allow incoming access. The default network includes some additional rules that override this one, allowing certain types of incoming connections.

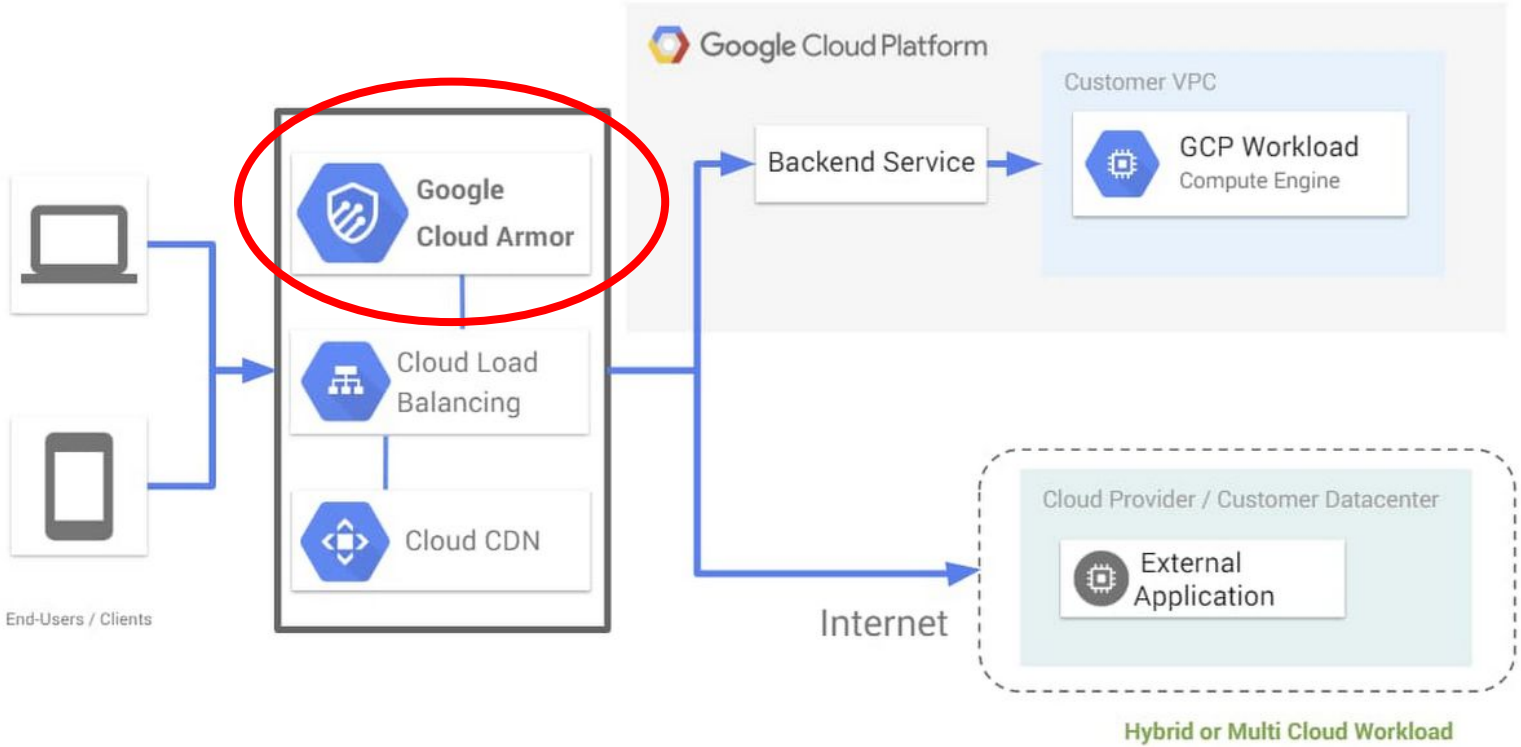
The implied rules cannot be removed, but they have the lowest possible priorities. You can create rules that override them as long as your rules have higher priorities (priority numbers less than 65535).
Because **deny** rules take precedence over **allow** rules of the same priority, an ingress allow rule with a priority of 65535 never takes effect.

Web Application Firewalls

- A Web Application Firewall (WAF) helps protect web applications by filtering and monitoring traffic between a web application and the Internet.
- Typically protects web applications from attacks such as cross-site forgery, cross-site-scripting (XSS), file inclusion, and SQL injection among others.
- A WAF is a layer 7 protocol defense (OSI model), and is not designed to defend against all types of attacks.
- This method of attack mitigation is usually part of a suite of tools which together create a holistic defense against a range of attack vectors.

https://www.cloudflare.com/en-gb/learning/ddos/glossary/web-application-firewall-waf/

Deploying a Web Application Firewall (WAF)



Cloud Armor pre-configured rules

The following table contains a comprehensive list of preconfigured WAF rules that are available for use in a Google Cloud Armor security policy. The rule source is ModSecurity Core Rule Set 3.0.2 (CRS).

Google Cloud Armor rule name	ModSecurity rule name	Current status
sqli-stable	SQL injection	In sync with sqli-canary
sqli-canary	SQL injection	Latest
xss-stable	Cross-site scripting	In sync with xss-canary
xss-canary	Cross-site scripting	Latest
lfi-stable	Local file inclusion	In sync with lfi-canary
lfi-canary	Local file inclusion	Latest
rfi-stable	Remote file inclusion	In sync with rfi-canary
rfi-canary	Remote file inclusion	Latest
rce-stable	Remote code execution	In sync with rce-canary
rce-canary	Remote code execution	Latest

https://cloud.google.com/armor/docs/rule-tuning

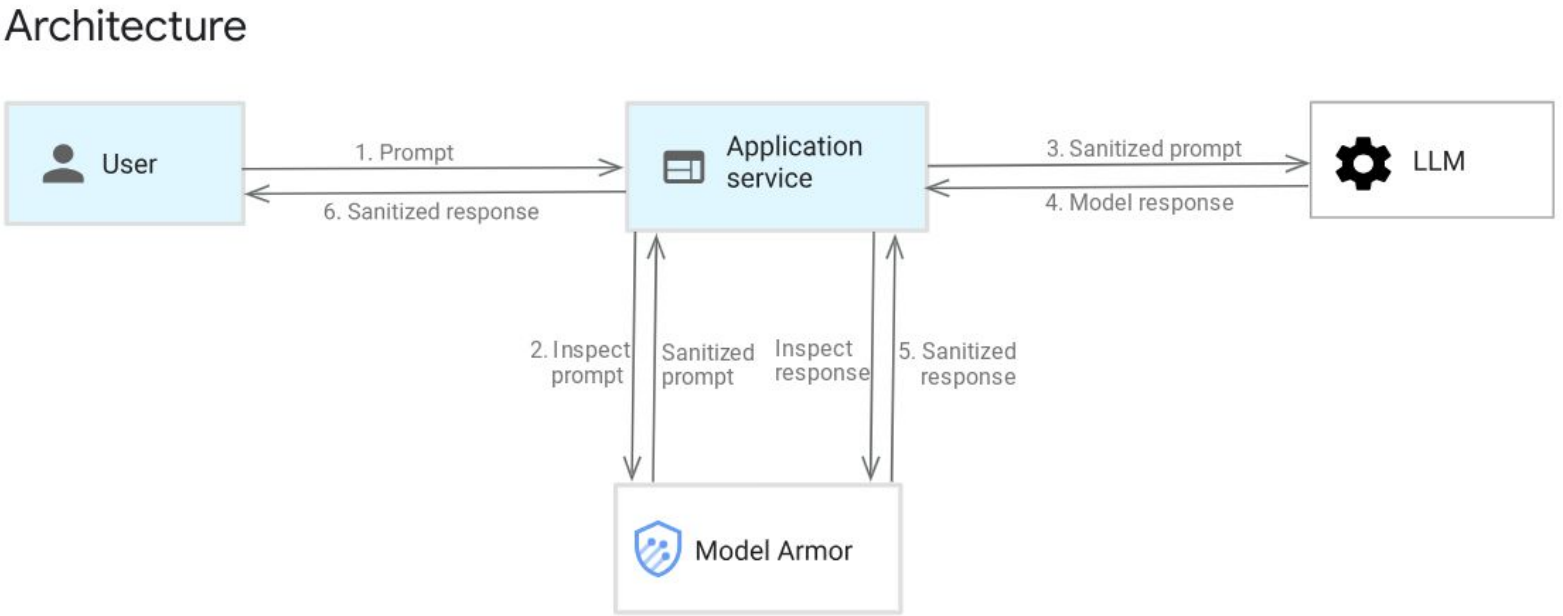
Model Armor

Model Armor is a fully managed Google Cloud service that enhances the security and safety of AI applications by screening LLM prompts and responses for various security and safety risks. Model Armor offers a number of features, including the following:

- **Model-independent and cloud-independent**: Model Armor is designed to support any model on any cloud platform. That includes multi-cloud and multi-model scenarios to choose the best AI solutions for your specific needs.
- **Centralized management and enforcement**: Model Armor enables centralized management and enforcement of security and safety policies.
- **Public REST APIs**: Model Armor provides a public REST API, allowing you to integrate prompt and response screening directly into your applications. This API-based approach supports various deployment scenarios.
- **Role-based access control (RBAC)**: Model Armor incorporates role-based access control (RBAC) to manage access and permissions within the service so that different user roles have appropriate levels of control and visibility.
- **Regional endpoints**: Model Armor's API is exposed using regional endpoints, providing low latency.
- **Multiple regions**: Model Armor is accessible throughout various regions in the United States and Europe.
- **Integration with Security Command Center**: Model Armor is integrated with Security Command Center, you to view the findings in the Security Command Center dashboard and identify violations and remediate them from the source.

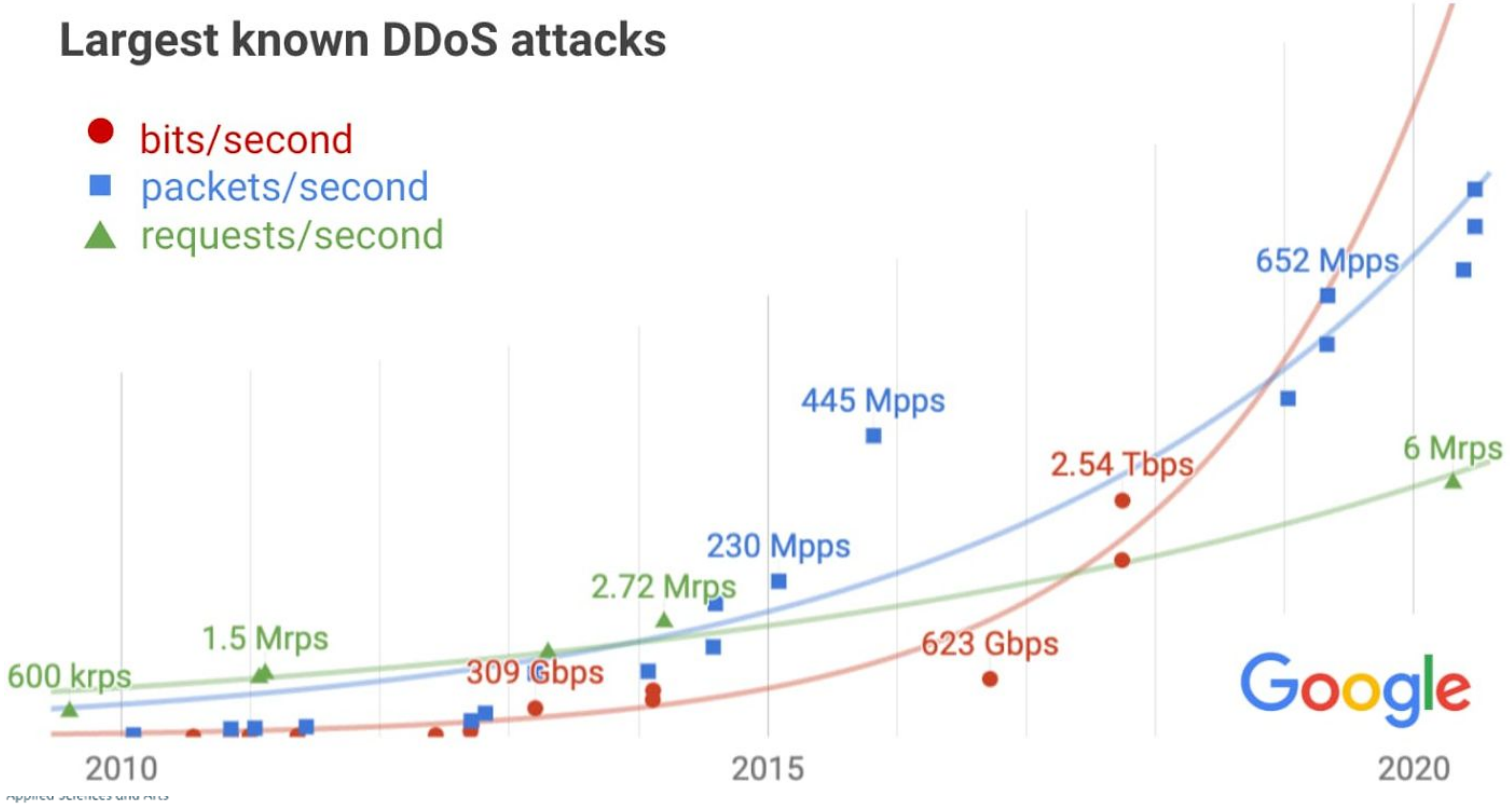
https://cloud.google.com/security-command-center/docs/model-armor-overview

Model Armor



https://cloud.google.com/security-command-center/docs/model-armor-overview

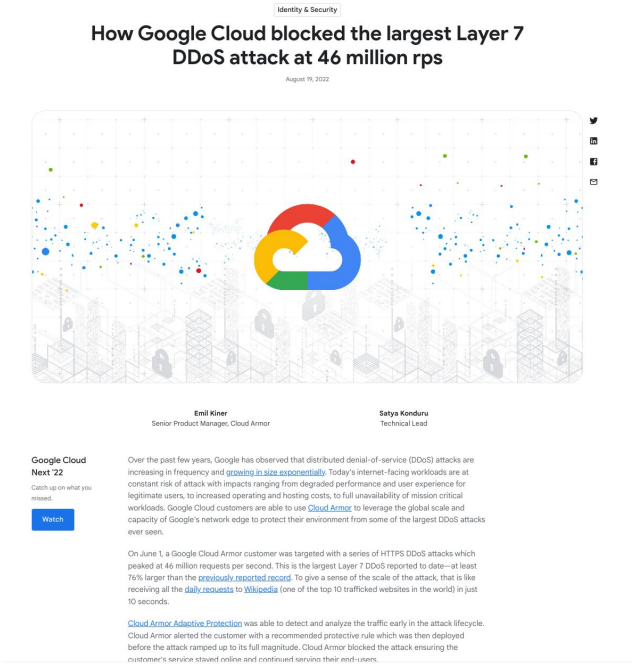
Exponential growth in DDoS attack volumes



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See https://cloud.google.com/blog/products/identity-security/identifying-and-protecting-against-the-largest-ddos-attacks

DDoS attack at 46 million rps (Aug 2022)



Identity & Security

How Google Cloud blocked the largest Layer 7 DDoS attack at 46 million rps

August 19, 2022

Emil Kiner
Senior Product Manager, Cloud Armor

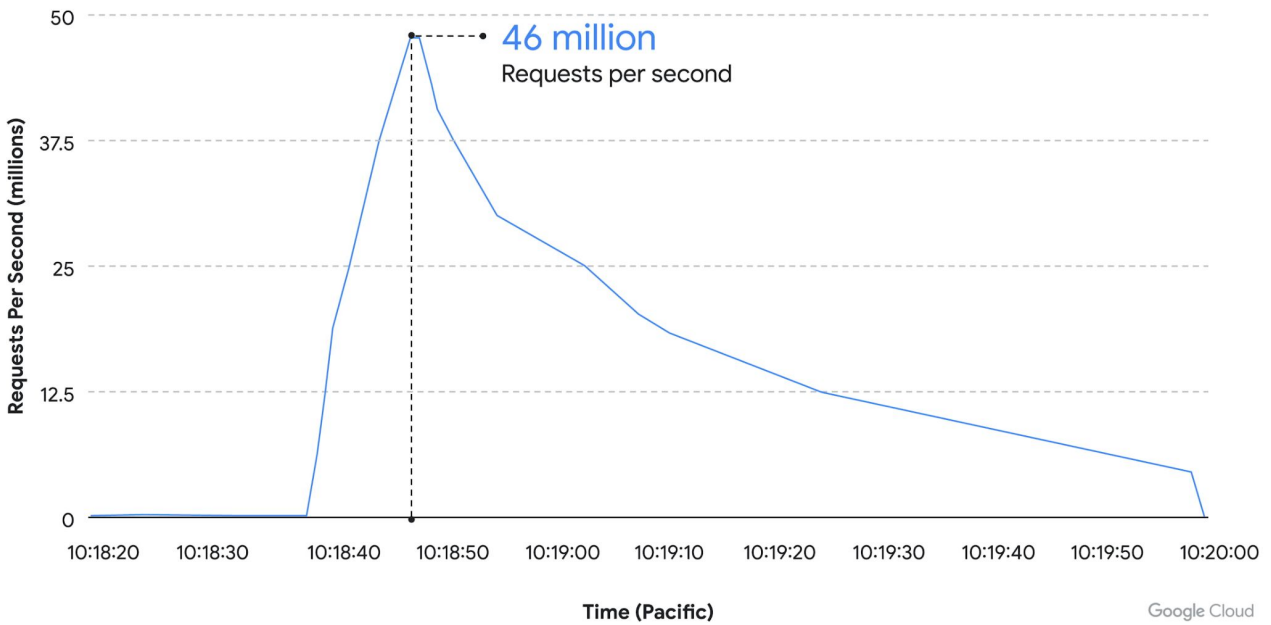
Satya Konduru
Technical Lead

Google Cloud Next '22
Catch up on what you missed.
Watch

Over the past few years, Google has observed that distributed denial-of-service (DDoS) attacks are increasing in frequency and growing in size exponentially. Today's internet-facing workloads are at constant risk of attack with impacts ranging from degraded performance and user experience for legitimate users, to increased operating and hosting costs, to full unavailability of mission critical workloads. Google Cloud customers are able to use Cloud Armor to leverage the global scale and capacity of Google's network edge to protect their environment from some of the largest DDoS attacks ever seen.

On June 1, a Google Cloud Armor customer was targeted with a series of HTTPS DDoS attacks which peaked at 46 million requests per second. This is the largest Layer 7 DDoS reported to date—at least 76% larger than the previously reported record. To give a sense of the scale of the attack, that is like receiving all the daily requests to Wikipedia (one of the top 10 trafficked websites in the world) in just 10 seconds.

Cloud Armor Adaptive Protection was able to detect and analyze the traffic early in the attack lifecycle. Cloud Armor alerted the customer with a recommended protective rule which was then deployed before the attack ramped up to its full magnitude. Cloud Armor blocked the attack ensuring the customer's service stayed online and continued serving their end-users.



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https://cloud.google.com/blog/products/identity-security/how-google-cloud-blocked-largest-layer-7-ddos-attack-at-46-million-rps

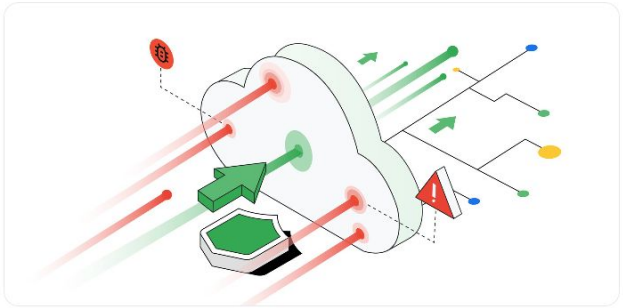
[How it works: The novel HTTP/2 'Rapid Reset' DDoS attack]

DDoS attack at 398 million rps (Oct 2023)

Security & Identity

Google mitigated the largest DDoS attack to date, peaking above 398 million rps

October 10, 2023



Emil Kiner
Senior Product Manager, Cloud Armor

Tim April
Security Reliability Engineer

The attack used a novel technique, HTTP/2 Rapid Reset, based on stream multiplexing

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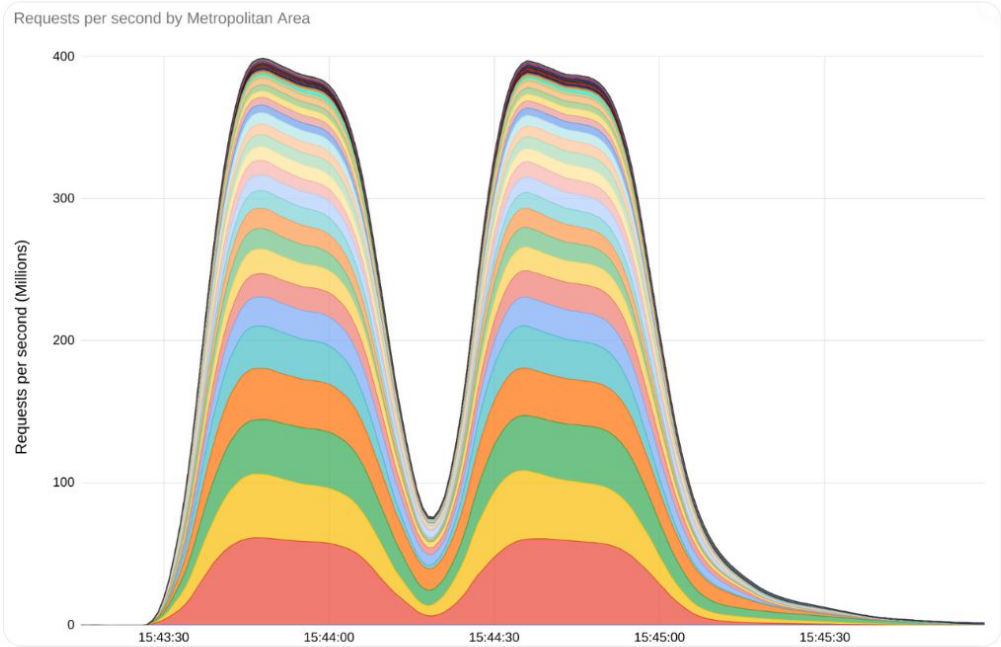
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Over the last few years, Google's DDoS Response Team has observed the trend that distributed denial-of-service (DDoS) attacks are increasing exponentially in size. Last year, we blocked the largest DDoS attack recorded at the time. This August, we stopped an even larger DDoS attack — 7½ times larger — that also used new techniques to try to disrupt websites and Internet services.

This new series of DDoS attacks reached a peak of 398 million requests per second (rps), and relied on a novel HTTP/2 "Rapid Reset" technique based on stream multiplexing that has affected multiple Internet infrastructure companies. By contrast, last year's largest-recorded DDoS attack peaked at 46 million rps.

For a sense of scale, this two minute attack generated more requests than the total number of article views reported by Wikipedia during the entire month of September 2023.



Google mitigated a DDoS attack which peaked at 398 million requests per second

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https://cloud.google.com/blog/products/identity-security/google-cloud-mitigated-largest-ddos-attack-peaking-above-398-million-rps/

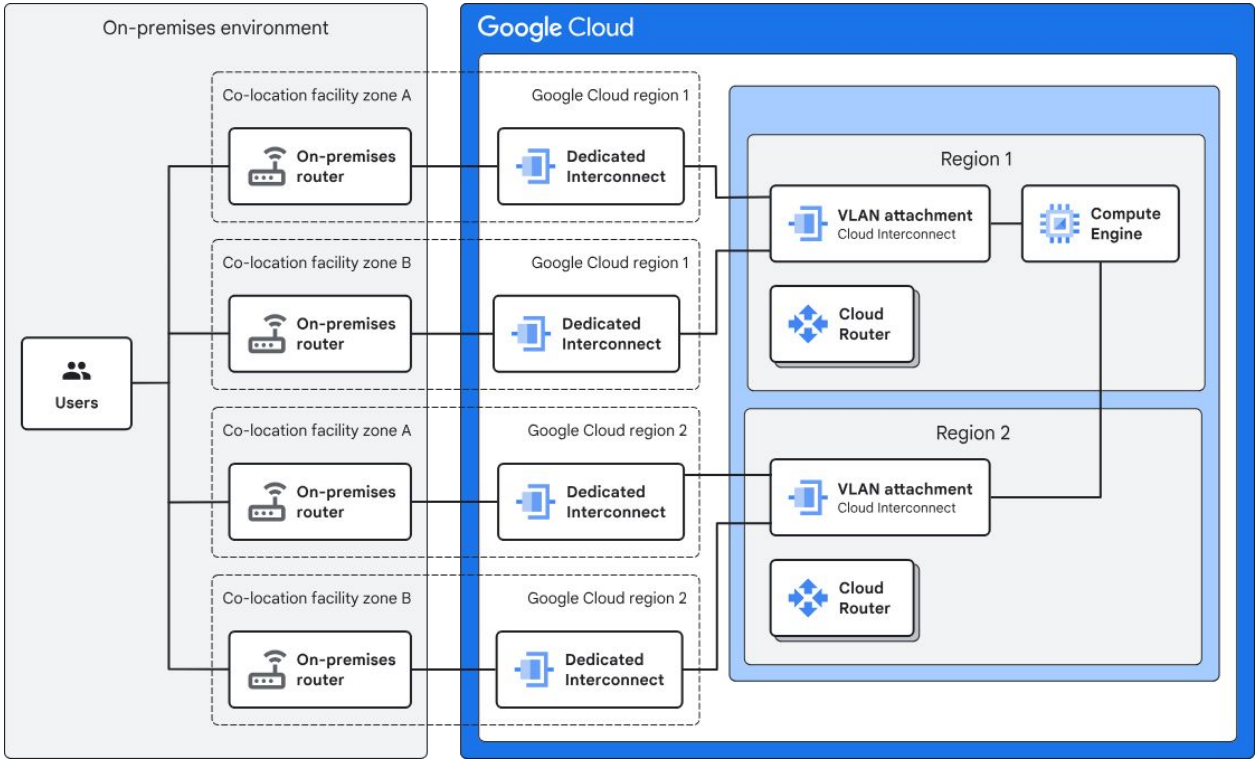
Connecting to the On Prem Network

ENCRYPTION IN TRANSIT WAS ADDRESSED IN LECTURE #3

- Cloud VPN
  - Cloud VPN enables connection of a VPC network to a physical, on-premises network or other cloud provider using a secure virtual private network.

- Cloud Interconnect
  - Cloud Interconnect allows connection of a VPC network to an on-premises network using a high speed physical connection
  - This can be a Dedicated interconnect or Partner interconnect (as also covered previously relating to encryption: see Encryption in Transit)

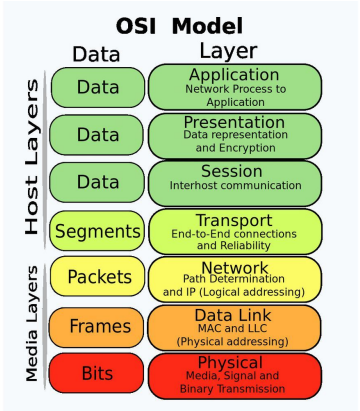
Dedicated Interconnect Connection Structure



https://docs.cloud.google.com/architecture/blueprints/security-foundations/networking

MACsec for Cloud Interconnect

- MACsec for Cloud Interconnect secures traffic on Cloud Interconnect connections, specifically between the on-premises router and Google's edge routers.
- MACsec for Cloud Interconnect uses IEEE standard 802.1AE Media Access Control Security (MACsec) to encrypt traffic between your on-premises router and Google's edge routers.



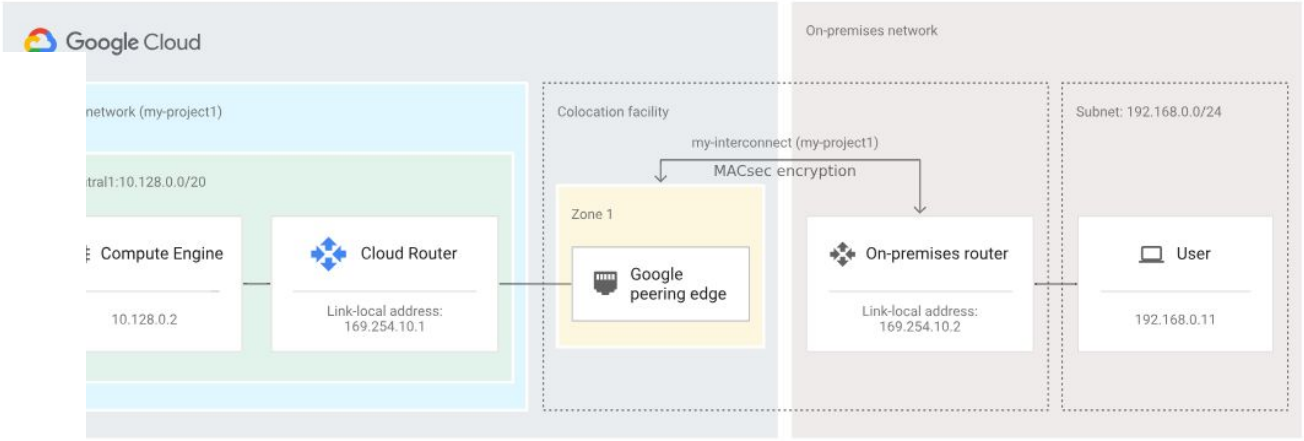


Figure 1. MACsec encrypts traffic on Dedicated Interconnect between Google's peering edge router and an on-premises router (click to enlarge).

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See: https://cloud.google.com/network-connectivity/docs/interconnect/concepts/macsec-overview

GCP Network Billing

- GCP, similar to other cloud providers, bills clients for egress traffic only.
- **Egress** traffic is considered as traffic to the Internet, or from one region to another (in the same network), or between zones within a region.
- GCP does not bill clients for ingress traffic.
- **Ingress** traffic includes VM-to-VM traffic in a single zone (same region, network), and traffic to most GCP services.

Quiz 10

- What is the difference between Web Application Firewalls and more general Firewalls?
- What is the difference between internal and external IP addresses in a VPC?
- What is an RFC1918 IP address?



GCP Networking and Security best practices

- Use VPC to define your network
- Manage traffic with firewall rules
- Limit external access
- Centralize network control
- Connect your enterprise network
- Secure your apps and data